

Wheel Time

6/24/26

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Wheel Time

- Real Time Application Launcher
 - Most launchers are keyboard focused, when playing games if you want quick access to say screenshotting, a color picker, or perhaps some macro/hook this will make your life faster
 - General computer users will also be helped if they have trouble remembering hotkeys, such an interface can easily provide actions with only 1 hotkey to remember instead

Project Scope

- Have a functional launcher
- Be able to run macros from the interface
- Some form of real time low latency interface

Sprint 1

Sprint Goal: Create a basic Windows launcher prototype with a simple interface.

- User Stories
 - As a computer user, I want to be able to launch programs and execute scripts quickly so I can run things that I use often without having to exit to desktop or type anything in a terminal - High Priority - 13 Story Points
 - As a gamer, I want the interface to not obscure the entire screen so I can still see what is happening while performing actions - Low Priority - 2 Story Points
- Spikes
 - Learn how Windows API works and interacts with graphics elements
 - Research app launching in C++
 - Research GUI/window options
- Infrastructure tasks
 - Create GitHub repo
 - Add README/build steps
 - Create basic launcher window
 - Set up C++/CMake project

Sprint 2

Sprint Goal: Allow the launcher to open and close quickly using a hotkey and run basic actions.

- User Stories
 - As a gamer, I want to be able to bring up the interface quickly so I don't have to interrupt my playing to access an action - Medium-High Priority - 5 Story Points
 - As a gamer, I want to be able to exit the interface quickly so that if something happens while I have it up I can react to it quickly - Medium-High Priority - 3 Story Points
 - As a computer user, I want to be able to run macros/hotkeys from a visual interface so I don't have to remember all of them - High Priority - 3 Story Points
- Spikes
 - Research Windows API global hotkeys.
 - Research background tray app vs background service.
 - Research simple macro/script execution.
- Tasks
 - Implement hotkey to open launcher.
 - Implement Escape or another key to close launcher.
 - Return focus to previous app after closing.
 - Add test script/macro action.
 - Test open/close behavior while another app is focused.

Sprint 3

Sprint Goal: Allow users to configure launcher actions without editing code.

- User Stories

- As a general computer user, I want to have a settings menu so I can easily control how the app works and add new buttons without having to modify a text file - Medium-High Priority - 13 Story Points
- As a user, I want to customize my launcher actions so I can choose the programs/scripts I use most often - Medium-High Priority -

- Spikes

- Research simple configurable GUI patterns.
- Research saving settings to a local file.
- Research action configuration format.

- Tasks

- Create basic settings menu.
- Add ability to edit action name.
- Add ability to edit action path/command.
- Save settings to a local settings file.
- Load saved actions when app starts.

Sprint 4

Sprint Goal: Polish the launcher, test performance, fix bugs, and prepare the final demo.

- User Stories

- As a gamer I want the interface to be lightweight so I can run it without worrying about lowering my FPS - Medium Priority
- As a user, I want the launcher to behave reliably so I can trust it during normal computer use or gaming - High Priority

- Spikes

- Research lightweight performance monitoring tools.
- Research common causes of UI lag or high CPU usage.

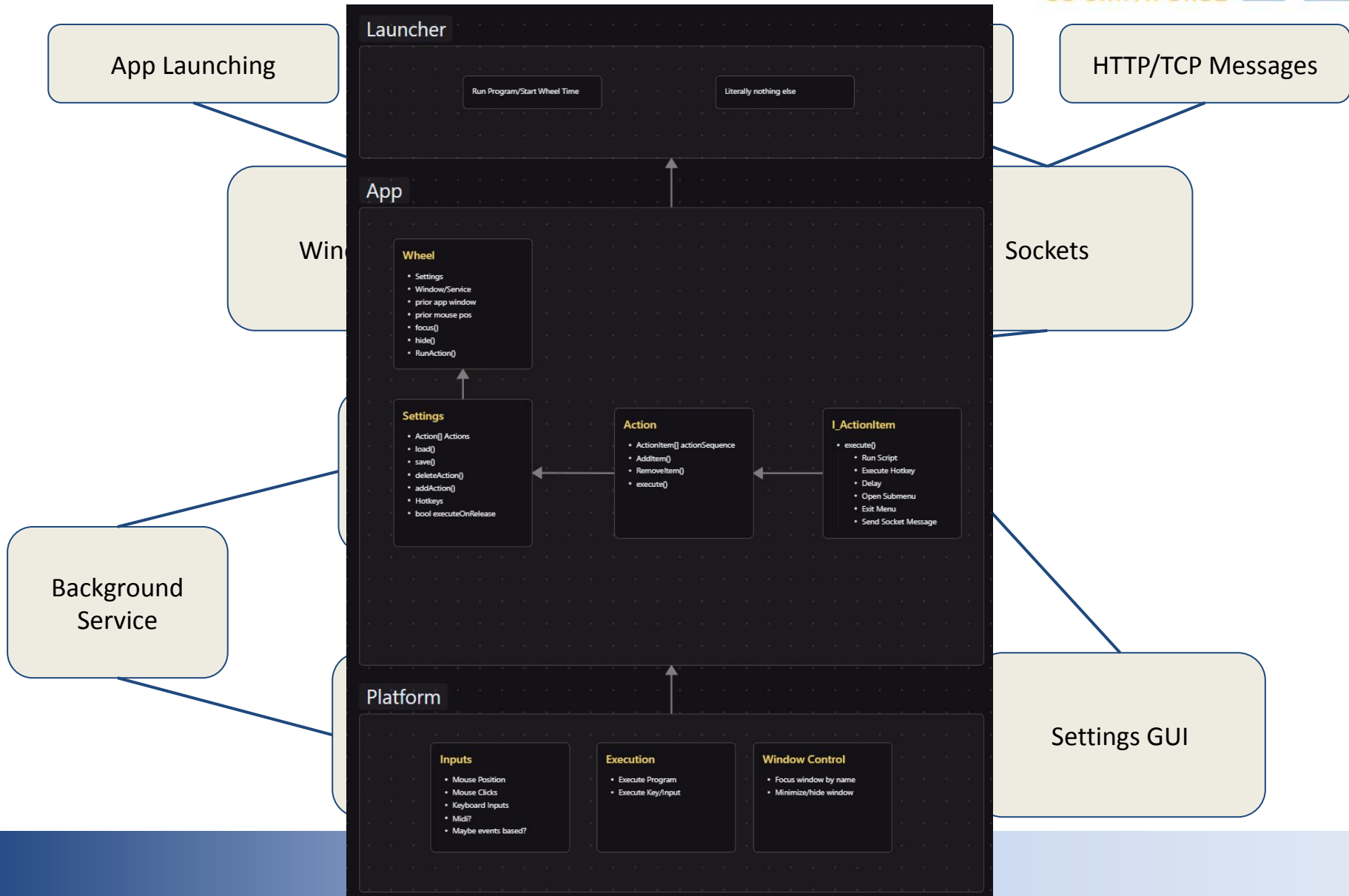
- Tasks

- Measure CPU and memory usage while idle.
- Measure CPU and memory usage while launcher is open.
- Test launcher with a fullscreen or game-like application.
- Fix open/close bugs.
- Clean up UI.
- Update README and demo instructions.
- Prepare final demo script.

Stretch / Backlog

- Linux support
- Search feature
- LRU/recent actions
- Recursive nesting/action groups
- Custom shapes/themes
- Render pipeline injection
- Discord/OBS integrations

Architecture



Technologies

Language: C++

Core Libraries: Windows API

Development Environment: Cursor, VSCode,
Git/Github

Other: Performance monitor(Afterburner)

If Extra Time: App Integrations(Discord, OBS,
etc.), Linux Libs

Challenges/Risks

Challenge 1: Unfamiliar Windows API

Challenge 2: Unfamiliar tools/build system

Challenge 3: Time constraints

Challenge 4: Low latency/performance issues

Minimum Viable Product (MVP)

Our MVP would be an app launcher that you can pull up quickly in a game and returns control to the game quickly.

- As a computer user, I want to be able to launch programs and execute scripts quickly so I can run things that I use often without having to exit to desktop or type anything in a terminal
- As a gamer, I want to be able to bring up the interface quickly so I don't have to interrupt my playing to access an action
- As a gamer, I want to be able to exit the interface quickly so that if something happens while I have it up I can react to it quickly

Acceptance Criteria for MVP

The MVP is complete when:

- User can open the launcher with a hotkey.
- Launcher displays a visual list of actions.
- User can launch at least one external program or script.
- User can close the launcher quickly with a key or click.
- Launcher returns control back to the previous application.
- App builds and runs on at least two team members' Windows machines.
- Basic usage is documented in the README.